

PROGRAM A: FCFS DISK SCHEDULING

C PROGRAM

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
int main()
```

```
{
```

```
    int req[20], n, head, i;
```

```
    int seek = 0;
```

```
    printf("Enter Number of Requests: ");
```

```
    scanf("%d", &n);
```

```
    printf("Enter Request Queue:\n");
```

```
    for (i = 0; i < n; i++)
```

```
    {
```

```
        scanf("%d", &req[i]);
```

```
    }
```

```
    printf("Enter Initial Head Position: ");
```

```
    scanf("%d", &head);
```

```
    for (i = 0; i < n; i++)
```

```
    {
```

```
        seek += abs(req[i] - head);
```

```
        head = req[i];
```

```
    }
```

```
    printf("Total Head Movement = %d\n", seek);
```

```
    return 0;
```

}

SHELL SCRIPT

```
GNU nano 8.7.1
#!/bin/bash

queue=(98 183 37 122 14 124 65 67)
head=53
seek=0

for req in "${queue[@]}"
do
    diff=$((req - head))

    if [ $diff -lt 0 ]
    then
        diff=$((-diff))
    fi

    seek=$((seek + diff))
    head=$req
done

echo "Total Head Movement = $seek"
```

Output

```
(ganesh@LAPTOP-R3RDH9NG)~$ nano fcfs.sh
(ganesh@LAPTOP-R3RDH9NG)~$ nano fcfsdisk.sh
(ganesh@LAPTOP-R3RDH9NG)~$ chmod +x fcfsdisk.sh
(ganesh@LAPTOP-R3RDH9NG)~$ bash fcfsdisk.sh
Total Head Movement = 640
```